

4(a) In SHM, frequency refers to the number of oscillations per unit time.
Angular frequency is the frequency $\times 2\pi$.

(b)(i) At the highest point of oscillation, the spring is stretched by 0.24 m.

Hence,

$$\begin{aligned} F_{net} &= mg - T \\ &= 4.0 \times 9.81 - 109 \times (0.36 - 0.12) \\ &= 13 \text{ N} \end{aligned}$$

(ii)

$$a_{\max} = \omega^2 x_0$$

$$\frac{F_{net}}{m} = \omega^2 x_0$$

$$\frac{13.1}{4.0} = \omega^2 \times 0.12$$

$$\omega = 5.2 \text{ rad s}^{-1}$$

Since you don't know where zero GPE is, do this column last

(iii)

	Kinetic Energy / J	Gravitational Potential Energy / J	Elastic Potential Energy / J	Total Energy / J
highest point	0	-3.1	$\frac{1}{2} (109)(0.24^2)$ = 3.1	0
equilibrium position	$\frac{1}{2} m\omega^2 x_0^2$ = $\frac{1}{2} (4.0)(5.22^2)(0.12^2)$ = 0.79	-7.9	$\frac{1}{2} (109)(0.36^2)$ = 7.1	0
lowest point	0	-12.6	$\frac{1}{2} (109)(0.48^2)$ = 12.6	0